Jon Wick

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Summary

Motivated molecular, cellular, and computational biologist with a B.S. in Molecular and Cellular Biology and a Computer Science minor. Over two years in the Lee lab at WWU, developed custom Python and R pipelines for genome-wide differential expression and small RNA analysis, and processed high-throughput sequencing data in Unix/HPC environments with careful quality control and reproducible, version-controlled workflows. Additional hands-on experience in western blotting, PCR, RT-PCR, and aseptic technique, including coursework on ER-associated protein degradation in *C. elegans* and an RNA interference pathway in *Tetrahymena thermophila*. Eager to apply these skills toward identifying and treating oncogenic targets.

Skills and Technologies

Molecular Biology & Techniques: Western Blot, PCR, Reverse Transcriptase PCR, End-Point PCR, NanoDrop Spectrophotometry, Aseptic Technique, Media/Buffer Preparation, Cell Culture (*Tetrahymena thermophila*, *C. elegans*)

Bioinformatics & Quantitative Analysis: RNA-seq Processing, Differential Expression (DESeq2), Genome Remapping, Genomic Data Analysis, bedtools, samtools, Statistical Analysis, Data Visualization (ggplot2, matplotlib)

Software & Tools: ImageJ, SnapGene, Benchling (electronic lab notebook), Git, Linux, Command Line Interface, SSH, Jupyter Notebook, Microsoft Office/Libre Office

Programming: Python, R, Bash, SQL, HPC Job Scheduling

Research Experience

Undergraduate Researcher, Lee Lab, WWU – Bellingham, WA June 2024 - present
Dr. Suzanne Lee and Dr. Dan Pollard

- Investigated the role of small non-coding RNAs in an RNA interference pathway that maintains genome integrity in *Tetrahymena thermophila*, integrating molecular and computational approaches to gene silencing.
- Characterized RNAi pathway components by Western blot and maintained live *T. thermophila* cultures under sterile, aseptic conditions.
- Developed custom Python and R pipelines for genome-wide sRNA biogenesis locus identification and differential expression analysis of high-throughput sequencing data.
- Prepared media, buffers, and reagents; maintained experimental records and electronic lab notebooks in Benchling.
- Presented findings at lab meetings, journal clubs, and the WWU SURP poster session.

Biotech Research Intern, Rain Incubator – Tacoma, WA May 2022 - December 2022
Dr. Jennifer McKee-Johnson

- Designed and executed pilot PCR-based experiments for tracking European Green Crab migration.
- Gained foundational experience including hypothesis development, literature review, and experimental planning.

Coursework Research Projects

ERAD Research Project – Advanced Molecular and Cell w/ Lab, BIOL 487, WWU Spring 2025
Dr. Lina Dahlberg

- Investigated ubiquitin-mediated ER-associated protein degradation (ERAD) via the E2 ubiquitin ligase *ubc-6* in *C. elegans* — mechanistically relevant to targeted protein degradation, which recruits the same E2/E3 ligase machinery to route substrates to the proteasome or lysosome.
- Quantified misfolded protein accumulation in *ubc-6* knockout lines by Western blot, Reverse Transcriptase PCR, and end-point PCR to compare protein turnover across genotypes.
- Co-authored a poster presented at the Northwest Worm Meeting and co-wrote a micropublication on the findings.

Genomic Data Analysis – BIOL 477, WWU Winter 2025
Dr. Matthew Zinkgraf

- Conducted mRNA differential expression analysis on *Tetrahymena thermophila* datasets using DESeq2 to identify differentially expressed genes across experimental conditions.
- Processed and analyzed genomic data in a Unix environment using Bash, Python, and high-performance computing resources.
- Generated statistical analyses and publication-quality data visualizations; communicated findings through a written research paper and oral presentation.

Education

B.S. Molecular and Cellular Biology, Computer Science minor September 2023-June 2026
Western Washington University – Bellingham, WA

- **Honors:** Outstanding Graduate, WWU Dept of Biology, 2026
- **Relevant Upper Division Coursework:** Biochemistry, Organic Chemistry, Immunobiology, Advanced Molecular and Cell Biology with Laboratory, Cellular and Molecular Biology, Genetics, Microbiology, Genomic Data Analysis, Biostatistics.

Tacoma Community College – Dual Enrollment (High School) September 2021-June 2023

Publications

Understanding the Leadership Structure and Mentoring Model of an Extracurricular Robotics Team: Key Findings from a Case Study March 2024

SIGCSE 2024: Proceedings of the 55th ACM Technical Symposium on Computer Science Education V. 2
Hitender Oswal, ***Jon Wick***, Seth Tandon, Ashley Brewster, Sushil K Oswal

DOI: 10.1145/3626253.3635603

Posters

Re-identification of the genomic origins of small RNAs in the model organism *Tetrahymena thermophila* and characterization of those dependent on genome protective *RDN2*
2025 SURP poster session, October 2025.
Jon Wick, Noah Haight, Hannah Thorp, Dan Pollard, and Suzanne R. Lee

The Role of E2 Ligases ubc-6 and 7 in ERAD in C. elegans
Northwest Worm Meeting, June 2025.
Summer Simpson, Abby Mellin, Dylan Wilding, Natalie Patch, **Jon Wick**, Bree Nelson, Hannah Thorp, Caelin Sickler, and Lina Dahlberg

Remapping previously sequenced 23-24 nucleotide sRNAs to the 2020 Tetrahymena thermophila Genome
WWU SURP poster session, September 2024.
Jon Wick, Noah Haight, Suzanne R. Lee, Dan Pollard

Awards

Outstanding Graduate, WWU Dept of Biology, 2026 | Declan Barron Memorial Scholarship, 2025 | WWU Biology Summer Undergraduate Research Program Award, 2025

Additional Experience

Club Lead, Molecular Bioscience Club, WWU – Bellingham, WA August 2025 - June 2026
Organized research-opportunity and professional-development programming for students of all course standings, and helped students secure research lab placements.

Computing Education Research, Sushil K. Oswal, UW – Tacoma, WA February 2023 - May 2024
Co-authored and published a paper at the ACM SIGCSE conference on mentorship structures in STEM education; performed qualitative analysis on interview and survey responses.

Programming Lead, SOTAbots FIRST Team 2557 – Tacoma, WA September 2019 - June 2023
Developed production code used at the 2022 FIRST Robotics World Championship semifinals and the 2023 district championship.